Preston Crowe

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Security Engineer with 5+ years experience building detectors, internal tools, and TI infrastructure.

[EXPERIENCE]

ExtraHop Networks — Seattle, WA
Security Engineer III, March 2024 – Present

- Built an Apache Iceberg indicator data lake serving analyst queries at 100M+ indicator scale; exported TI collections to an enrichment service powering detections for all customers.
- Developed the first ExtraHop MCP server for agentic enumeration of detections and network devices, winning the 2025 hackathon audience choice award.

Security Engineer II, March 2022 – March 2024

- Automated AWS traffic-mirroring environments to benchmark detections against breach and attack simulation platforms; added coverage for 3,000+ Mandiant Security Validation (MSV) actions.
- Authored internal MSV testing guide and provided guidance for sales engineers in customer engagements.

Security Engineer I, December 2020 – March 2022

- Shipped detections within hours of disclosure for critical vulnerabilities (e.g., Log4Shell, SUPERNOVA).
- Developed detection strategies for red team tools (Redline, BloodHound, PetitPotam, Farmer); built pcap manipulation and automated traffic replay tooling with threat research and performance teams.
- Automated CVE triage for network vulnerabilities, eliminating a weekly one-hour meeting for a ~10 person threat research team (520 researcher-hours per year).

Baffin Bay Networks — Seattle, WA
Internship, Product Development – June 2019 – September 2019

- Built an internal dashboard to monitor a honeypot network hosted across dozens of cloud providers.
- Developed an API for Baffin Bay's Threat Insight platform to provide context enrichment for IPs.

F5 Networks — Seattle, WA
Internship, F5 Labs – June 2018 – September 2018

- Worked with a team of security researchers to develop a vulnerability data pipeline.
- Built a threat map with React and D3.js to visualize suspicious traffic in real time.
- Generated data visualizations and animations for a Black Hat 2018 talk.

NASA AMES Research Center — Mountain View, CA
Internship, OpenMCT – June 2017 – September 2017

- Implemented JSON import/export for workspaces in an open-source mission control application.

[SKILLS]

- **Detection & Response:** detection engineering, incident response, NDR, MITRE ATT&CK
- **Threat Intelligence:** TI collection and processing pipelines, exploit monitoring, OSINT
- **Threat Research:** exploit reproduction, red team tool analysis, vulnerability triage
- **Traffic Analysis:** PCAP analysis (Wireshark), traffic mirroring and replay (scapy / tcpreplay)
- **Data & AI Engineering:** Apache Iceberg (S3 Tables), Athena, MCP, agentic tooling, data lakes
- **Languages & Frameworks:** Python, Go, JavaScript, SQL, Java, C, React, Vue.js, D3.js
- **Cloud & Infrastructure:** AWS, Linux, Docker, Terraform, Ansible, GitLab CI/CD
- **Interests:** SDR, ESP32, LoRa, 3D printing, data hoarding, homelab, lockpicking

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[EDUCATION]

University of Washington — Seattle, WA
Sep 2016 – Jun 2020
B.S. Computer Science, Dean's List, 3.71 GPA

[PUBLICATIONS]

- F5 Networks, 2018: *Breaking Down the Door to Emergency Services through Cellular IoT Gateways*
- ExtraHop, 2021: *PetitPotam: Expanding NTLM Relay Attacks*